

Project reference **10 Ashley way** Document date **Mon, May 5, 2025**

Your heat pump proposal

Customer details

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Your heat pump proposal

Contents

1 | **About your project** Your installer About MCS

Your home

- In depth technical information about the proposed installation
- Heat loss & system design
- Your air source heat pump
- Your thermal storage
- Annual system performance estimates

2 | **Proposed upgrades**

Key information about your new air source heat pump system

- Heat pump location
- Room heat loss & current heat emitters Heat emitter upgrades

3 | **In detail**

4 | **Things to know** MCS: Key facts



1 | **About your project**

Your installer

About us \${company.business_description}



Project reference **10 Ashley way** Document date **Mon, May 5, 2025**

About MCS

Working with industry, MCS (the Microgeneration Certification Scheme) **sets, defines and maintains the standards for low-carbon energy technology products, contractors and their installations.** This includes heat pumps, solar, biomass, small



wind and battery storage.

MCS is a mark of quality. Membership of MCS demonstrates adherence to these recognised industry standards, highlighting quality and competency.



You'll find out more about some of these standards in the **MCS: Key Facts** sections of this document. You'll also see a **MCS Performance Comparison** within your annual performance estimate section. This is an industry standard energy usage estimate for your new heat pump system based on figures from your most recent Energy performance Certificate (EPC).

All MCS accredited installers are required to provide you with this estimate to ensure that you can make consistent comparisons between proposals. This document also contains additional performance estimates based on the detailed heat loss survey and final system design carried out for your home.

MCS aims to decarbonise heat and power in the UK's homes by **giving you confidence in home-grown energy**.

Visit the MCS website at **mcscertified.com** for more information



1 | About your project

Your home

Property age band

The age of your home is a good indicator of how well-insulated it is likely to be and allows us to build a picture of its fabrics and thermal performance

Floor area

As captured by your home survey. We use this to calculate your home's overall



Project reference **10 Ashley way**
Document date **Mon, May 5, 2025**

heat loss so that we can correctly size your heat pump and emitter upgrades

Number of bedrooms

As standard, your hot water system design is based on the number of bedrooms your home has. Any extra needs can be considered in your design

Number of current occupiers

Our estimates of your hot water usage are based on how many

people are
currently living in the home

In a Conservation Area

Conservation Area status may mean that there are extra restrictions around
locating a heat pump that we need to consider

In an Area of Outstanding Natural Beauty (AONB)

If your property is in an AONB there may be extra restrictions around locating
a heat pump that we need to consider

England and Wales: 1967- 1975

124 m²

4

4

No

No



Project reference **10 Ashley way** Document date **Mon, May 5, 2025**

2 | Proposed upgrades

Heat loss survey results Heating system design

Design inside temperature

Your new heating system is designed to maintain your living spaces at this temperature. Heat pumps work differently to boilers and it's most efficient to run them continuously throughout the day.

Design outside temperature

The heat loss calculation for your home uses this as an outdoor reference temperature. This is based on the typical coldest weather conditions at your location.

Peak home heat loss

This is how much heat your home loses during typical coldest weather, and needs to be matched by the heat pump's power to maintain its inside temperature. Your home loses much less heat than this most of the year.

22°C Peak flow temperature

This is the temperature that your new heating system is designed to send to your emitters (whether radiators or underfloor heating) when your home is at peak heat loss. This new flow temperature is probably lower than you're used to from your existing system, but don't worry!

-2.20°C Proposed new emitters

Designing at lower flow temperatures creates a more comfortable and efficient heating system, but it can mean that

some of your existing radiators or underfloor systems need to be upgraded or supplemented.

7.6kW SCoP

Seasonal Coefficient of Performance is a measure of efficiency, and means that on average for every 1kW of electricity it uses, it will generate \${{}} of heat for your home.

50°C 3.6

Project reference **10 Ashley way** Document date **Mon, May 5, 2025**

2 | Proposed upgrades

Your air source heat pump

Make Daikin

Model Altherma Monobloc 9kW

About your heat pump

Peak heat output

The real-world output of a heat pump isn't always the same as what's on the

badge and depends on
your home’s design
conditions

This is a measure of how
much noise the unit
makes at maximum power.
For comparison, modern
boilers make around 40-
60dB

Sound power level **9.1kW 62dB**

Project reference **10 Ashley way** Document date **Mon, May 5, 2025**

2 | **Proposed upgrades**

Your thermal storage

Make UK Cylinders

Model Elite 250L

About your thermal storage

Capacity

Your new thermal store has been sized according to you and your home's needs to ensure hot water is on tap when you need it

Immersion heater backup

The number and power output of any backup immersion heaters the unit features as standard

250 Litres 1 x 3kW

2 | **Proposed upgrades**

Annual system performance estimates

Performance estimates allow you to compare your existing heating system with the upgrade we’ve proposed, as well as with other proposals you might receive from other installers. See the Key Facts section of this document for more info

Home survey system performance estimate These estimates are based on the detailed heat loss survey we carried out in your home, and should provide the most accurate comparison between your old and new system

Property energy needs Energy required (kWh) Space heating 15454.866 Hot water 3820.64 **Total 19275.506**

CO₂ produced
(kg)
4,391
1,086
5,477

Existing heating system Fuel use (kWh) Fuel cost (£)

Fuel efficiency guideline \$ {calculations.perfor
Space heating 18,847 1,319 Hot water 4,659 326 **Total 23,507 1,645**

Proposed heat pump system

SCoP (Space heating) 3.6

SCoP (Hot water) 3.6

829
205
1,033

Space heating 4,293 1,159

Hot water 1,061 287

Total 5,354 1,446

Additional weekly immersion legionella cycle (if required) 136 26 37

Project reference **10 Ashley way** Document date **Mon, May 5, 2025**

3 | **In detail**

Heat pump location

Heat pump location *Coming soon*

Noise assessment result

This ensures that your heat pump won't disturb nearby neighbours

Full noise assessment calculation (to MCS standard: MIS 020)

Noise assessment position

This is an entry point into a neighbouring property used in the calculation

Sound power level

How much noise the heat pump when running at maximum power

Distance to assessment position	Rear of property below study window facing into 62dB
The distance from the heat pump to assessment position	
Adjustment for barriers	
Heat pump noise will be reduced if barriers (e.g. wall or fences) are present	6m
Calculated noise level	-10dB
A result below 42dB is considered a pass	
PASS	40.00dB

Project reference **10 Ashley way** Document date **Mon, May 5, 2025**

3 | In detail

Room heat loss & current heat emitters

These calculations are performed to the MCS standard MIS 3005 and according to BS EN12831

Room Existing heat emitters Heat loss & emitter performance

Area
m²

8.8
6.2
36.8

17.6
8.8
7.8

7.8
3.4
8.4
9.0
5.0
4.5

-
630x1400
630x1000
1600x500
630x1400
600x1600
1600x600
1600x600
-

51
47
79
60
53
49
74
59
83
61

Name Design temperature

Dimensions
h x w mm
-
-
-
-

Type Output at new flow temperature

W/m²
99
70

Heat loss Coverage of heat loss by emitters

Study 22 Underfloor 920 873 105% **Utility** 22 Underfloor 519 434 120% **Dining/kitchen** 22 Underfloor 3,074 1881 163% **Living** 22 Underfloor 1,840 830 222% **Hall** 22 Underfloor 920 697

132%

Bedroom guest small

Bedroom small with ensuite
22 Type T20 705 465 152% 22 Type T20 503 412 122%

Small ensuite 22 Type TR 195 167 117% **Bedroom master** 22 Type T20 705 624 113% **Bedroom guest** 22 Type T11 632 533 119% **Bath** 22 Type TR 234 413 57%

Bed & Ensuite 22 Type TR
234
Alternative

-
273 100%

Project reference **10 Ashley way** Document date **Mon, May 5, 2025**

3 | In detail

Heat emitter upgrades

These calculations are performed to the MCS standard MIS 3005 and according to BS EN12831

Room Heat emitters

Model	Status	Type	Dimensions h x w mm	Flow temperature °C	Output Watts
---	Existing	Underfloor	-	50	920
---	Existing	Underfloor	-	50	519
---	Existing	Underfloor	-	50	3,074
---	Existing	Underfloor	-	50	1,840

---	Existing	Underfloor	-	50	920
---	Existing	Type T20	630x1400	50	705
---	Existing	Type T20	630x1000	50	503
---	Existing	Type TR	1600x500	50	195
---	Existing	Type T20	630x1400	50	705
---	Existing	Type T11	600x1600	50	632
---	Existing	Type TR	1600x600	50	234
---	Existing	Type TR	1600x600	50	234
---	Existing	Alternative	-		-

Name Heat loss Make Coverage of heat loss

Watts

Study 873 --- 105% **Utility** 434 --- 120% **Dining/kitchen** 1881 --- 163% **Living** 830 --- 222% **Hall** 697 --- 132%

Bedroom guest small

Bedroom small with ensuite

465 --- 152% 412 --- 122%

Small ensuite 167 --- 117% **Bedroom master** 624 --- 113% **Bedroom guest** 533 --- 119% **Bath** 413 --- 57%

Bed & Ensuite 273 --- --- 100%

4 | Things to know

MCS: Key facts

About your performance estimate

Predicting the heat demand of a building (and therefore the performance and running costs of heating systems) is difficult to predict with certainty due to the variables discussed here. These variables apply to all types of heating systems, although the efficiency of heat pumps is more sensitive to good system design and installation. For these reasons your estimate is given as guidance only and should not be considered as a guarantee.

Seasonal Coefficient of Performance (SCoP) is derived from EU ErP labelling requirements, and is an indication of the anticipated efficiency of a heat pump over a whole year using standardised climate data. It is used to compare the relative performance of heat pumps under fixed conditions and indicates the units of total heat energy generated (output) for each unit of electricity consumed (input). As a guide, a heat pump with a MCS SCoP of 3 indicates that 3 kWh of heat energy would be generated for every 1 kWh of electrical energy it consumes over a 'standard' annual cycle.

Energy Performance Certificate (EPC) is a report produced in accordance with a methodology approved by the government. As with all such calculations, it relies on the accuracy of the information input. Some of this

information, such as the insulating and air tightness properties of the building may have to be assumed and this can affect the final figures significantly leading to uncertainty especially with irregular or unusual buildings.

We have identified 3 key types of factor that can affect how much energy a heating system will consume and how much energy it will deliver into a home. These are 'Fixed', 'Variable' and 'Random'. Most factors are common to all heating systems regardless of the type (e.g. oil, gas, solid fuel, heat pump etc.) although the degree of effect varies between different types of heating system as given in the following table.

The combined effect of these factors on energy consumption and the running costs makes overall predictions difficult, however an accuracy of $\pm 25\text{--}30\%$ would not be unreasonable in many instances. Under some conditions even this could be exceeded (e.g. regular opening of windows during heating periods). Therefore it is advised that when making choices based on mainly financial criteria (e.g. payback based on capital cost verses net benefits such as fuel savings and financial incentives) this variability is taken into account as it could extend paybacks well beyond the period of any incentives received, intended occupancy period, finance agreement period etc.

About your location assessment

This assessment is performed in accordance with the calculation methodology laid out in MCS 020: Planning Standard for Air Source Heat Pumps, including standard correction factors for distance reduction (Note 4), barriers between heat pump and assessment position (Note 5) and decibel correction (Note 7).

Multiple location assessments may need to be completed as part of a heat pump survey depending on the quantity of entry points into neighbouring buildings.

MCS 020 also requires the heat pump hardware to be certified in accordance with MCS 007 and installed by an MCS Contractor in accordance with MIS 3005.

Distance from property boundary means the horizontal distance of all parts of the heat pump to the property boundary. In order for the installation to qualify for permitted development rights this distance must be a minimum of one metre.

Assessment position means a position one metre external to the centre point of any door or window to a habitable room of a neighbouring property as measured perpendicular to the plane of the door or window.

